

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Jabaraj T (192511187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Selected Research Article

Article Citation Details

Title: *Machine Learning Based Diabetes Classification and Prediction for Healthcare Applications*

Authors: Umair Muneer Butt, Sukumar Letchmunan, Mubashir Ali, Fadratul Hafinaz Hassan, Anees Baqir, and Hafiz Husnain Raza Sherazi.

Journal: *Journal of Healthcare Engineering*

Year: 2021

Volume: 2021

Article ID: 9930985

DOI: 10.1155/2021/9930985

Publication Date: 1 October 2021.

Domain: Healthcare / Medical Diagnosis / IoT-based Health Monitoring

Machine Learning Problem: Diabetes classification and early diabetes prediction.

Dataset: PIMA Indian Diabetes Dataset.

TASK 1: ARTICLE SUMMARY

1(a) Domain and Specific ML Problem

The selected research article focuses on the application of Machine Learning in healthcare, particularly for diabetes classification, early identification and blood-glucose prediction.

Diabetes is a major healthcare problem because uncontrolled diabetes can contribute to serious complications. The authors aim to use machine learning to analyse medical variables and assist in identifying diabetes at an early stage.

The paper addresses two related machine learning problems:

1. Diabetes Classification – determining whether a person belongs to a diabetes category using classification algorithms.
2. Diabetes Prediction/Forecasting – predicting future blood glucose behaviour using historical patient information.

For classification, the authors use:

- Logistic Regression
- Random Forest
- Multilayer Perceptron (MLP)

For prediction, they use:

- Moving Average
- Linear Regression
- Long Short-Term Memory (LSTM)

The study also proposes a hypothetical IoT-based diabetes monitoring system in which Bluetooth Low Energy (BLE) sensors, smartphones, Apache Kafka and MongoDB can be used to collect and process patient information in real time.

1(b) Article Summary – 150–200 Words

Summary

The article proposes a machine learning-based framework for diabetes classification, early identification and blood-glucose prediction in healthcare. The authors identify the need for intelligent healthcare systems because conventional diabetes monitoring can require repeated medical visits and analysis of several patient parameters. Existing studies had explored individual machine learning algorithms, but the authors aimed to combine classification, prediction and real-time healthcare monitoring within a broader framework. The study uses the PIMA Indian Diabetes dataset containing 768 records and eight medical attributes. Logistic Regression, Random Forest and a fine-tuned Multilayer Perceptron are used for diabetes classification, while Moving Average, Linear Regression and Long Short-Term Memory (LSTM) are investigated for blood-glucose prediction. The fine-tuned MLP achieves 86.083% classification accuracy, while LSTM achieves 87.26% prediction accuracy. The study additionally proposes a hypothetical IoT architecture using BLE sensors, smartphones, Apache Kafka and MongoDB for real-time monitoring. The authors conclude that combining machine learning with IoT can support early diabetes identification and continuous monitoring, although broader datasets and further implementation are needed.

Word count: approximately 175 words.

The paper's stated objective is twofold: classification of diabetes into predefined categories and predictive analysis of diabetes using several machine learning approaches.

Research Gap Identified

From the literature review, the authors identify several limitations in previous studies:

- Some studies concentrated only on diabetes classification.
- Others concentrated only on diabetes prediction.
- Several studies did not provide state-of-the-art comparisons.
- Some studies used limited datasets.
- Real-time healthcare data processing was not adequately integrated with diabetes classification and prediction.

The paper's comparison table shows that the proposed study attempts to combine diabetes classification, diabetes prediction and real-time healthcare data analysis, whereas many earlier studies addressed only one or two of these aspects.

How the authors address the gap

The authors propose an integrated framework using:

Machine Learning + Diabetes Prediction + IoT Monitoring

The framework combines ML models with a hypothetical real-time monitoring architecture involving BLE sensors, smartphones, Apache Kafka and MongoDB.

TASK 2: METHODOLOGY ANALYSIS

2(a) ML Techniques Used

The article uses two major groups of machine learning techniques.

A. Diabetes Classification

Three algorithms are used:

1. Logistic Regression

Logistic Regression is used because the classification problem has categorical outcomes. It uses the sigmoid function to calculate the probability of a class.

The general form is:

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

The authors use Logistic Regression as a statistical classification baseline.

2. Random Forest

Random Forest is an ensemble learning technique consisting of multiple decision trees.

The algorithm:

1. Selects samples from the training data.
2. Builds individual classification trees.
3. Produces predictions from the trees.
4. Uses the combined predictions to determine the final classification.

The paper describes Random Forest as an ensemble in which multiple models contribute to the final decision.

3. Multilayer Perceptron (MLP)

The authors fine-tune an MLP for diabetes classification.

The architecture contains:

Input Layer

↓

Hidden Layer

↓

Output Layer

The input layer contains eight variables. Sigmoid activation is used, and backpropagation is used to update weights and reduce prediction error.

B. Diabetes Prediction

The authors use three approaches:

1. Moving Average

Historical observations are averaged to estimate future values.

2. Linear Regression

Linear Regression models the relationship between independent variables and the dependent variable.

3. Long Short-Term Memory (LSTM)

LSTM is a recurrent neural network architecture designed to capture patterns over time.

The authors selected LSTM because its memory mechanism can retain relevant information over longer sequences. The model uses gates to control which information is retained, updated and output.

Dataset Description

The paper uses the PIMA Indian Diabetes Dataset.

The dataset contains:

- 768 records
- Female patients
- Age greater than 21
- 8 input variables
- One target/output variable

The target variable, Outcome, has two possible values:

- 0 → Non-diabetic
- 1 → Diabetic

The authors state that the dataset is intended to predict whether a PIMA Indian woman has a chance of diabetes within the coming four years.

Dataset Features

No. Feature	Description
1 Pregnancies	Number of pregnancies
2 Glucose	Plasma glucose concentration
3 Blood Pressure	Blood pressure in mmHg
4 Skin Thickness	Triceps skinfold thickness
5 Insulin	Serum insulin level
6 BMI	Body Mass Index
7 Diabetes Pedigree Function	Diabetes-related hereditary indicator
8 Age	Age in years
9 Outcome	Diabetes class: 0 or 1

The paper gives the statistical descriptions and ranges of these variables in its dataset table.

Data Preparation

The paper uses missing-data handling as part of preprocessing.

The authors explain that their framework processes the data before applying the prediction models. The study uses:

- Missing-data handling
- Feature preparation
- Model training
- Model evaluation

The paper states that the classification and prediction models use 10-fold cross-validation for training and testing.

For the missing-data problem, the study considers several approaches, including:

- Fixed median
- Random normal
- Random uniform

The authors state that these methods were used to impute missing data and prepare the datasets for model building.

Training and Testing

The LSTM prediction model uses:

- 80% data for training
- 20% data for testing

The authors also fine-tune the number of LSTM units in the cell state to identify more important features and dependencies between diabetes parameters and the output.

For the classification/prediction evaluation, the paper also reports using 10-fold cross-validation.

2(b) Mapping to CO4 Topics

The assessment's CO4 covers machine learning techniques including Decision Trees, Reinforcement Learning and classification/decision-making.

The selected article connects to CO4 as follows:

CO4 Concept	Connection with Article
Inductive Learning	Models learn patterns from labelled patient records and use them to classify/predict unseen cases.
Statistical Learning	Logistic Regression is used for diabetes classification.
Classification	Logistic Regression, Random Forest and MLP classify diabetes.
Decision Making	Model outputs can support early diabetes identification and healthcare decisions.
Prediction	LSTM, Linear Regression and Moving Average are used for prediction.
Decision Trees	Random Forest is an ensemble containing decision trees, although the paper does not focus on a standalone Decision Tree.
Reinforcement Learning	Not directly implemented in this article.

Important point

The article does not implement Reinforcement Learning. Therefore, I would not claim that it demonstrates RL. It is mainly relevant to Inductive Learning, Statistical Learning, classification and decision-making, with Random Forest providing a connection to tree-based learning.

Methodological Strength

One major strength is the comparative evaluation of multiple machine learning algorithms.

Instead of relying on a single model, the authors compare Logistic Regression, Random Forest and MLP for classification. They also compare Moving Average, Linear Regression and LSTM for prediction.

This makes it possible to determine which model performs better under the same experimental setting.

Another strength is that the authors connect the ML models to a potential real-time IoT healthcare architecture, making the research more practically oriented.

Methodological Limitation

The major limitation is the limited size and scope of the dataset.

The PIMA dataset contains only 768 records and represents female patients from a specific population. Therefore, the results may not generalise to all populations.

The use of a single benchmark dataset also limits external validation.

Another limitation is that the proposed IoT healthcare system is hypothetical, rather than a fully deployed clinical system. The paper proposes the architecture but states that implementation of the Android application is future work.

TASK 3: RESULTS & CRITICAL EVALUATION

3(a) Key Results

Diabetes Classification Results

The paper compares three classifiers.

Algorithm	Accuracy (%)	Recall (%)	Precision (%)
Logistic Regression	73.05	72.7	73.0
Random Forest	77.4	75.7	76.9
Fine-tuned MLP	86.083	85.1	86.6

The fine-tuned MLP performs substantially better than the Logistic Regression and Random Forest baselines in the reported experiment.

Best Classification Model

Fine-tuned MLP

- Accuracy = 86.083%
- Recall = 85.1%
- Precision = 86.6%

The paper reports these values as the best classification performance.

Diabetes Prediction Results

For blood-glucose prediction, the paper compares:

- Moving Average
- Linear Regression
- Fine-tuned LSTM

Algorithm	Correlation (r)	RMSE	Accuracy (%)
Moving Average	0.710	42.946	75.4
Linear Regression	-0.071	82.592	72.7
Fine-tuned LSTM	0.999	2.285	87.26

The LSTM achieves the strongest reported performance, with a correlation coefficient of 0.999, RMSE of 2.285, and accuracy of 87.26%.

State-of-the-Art Comparison

For diabetes prediction, the paper compares the proposed LSTM with previous methods:

Algorithm	Accuracy (%)
J48	67.9
K-Means	70.5
Feed Forward Neural Network	71.8
RB-Bayes	72.9
Naïve Bayes	76.3
Artificial Neural Network	85.09
Proposed LSTM	87.26

Thus, the proposed LSTM has the highest reported accuracy among the listed approaches.

3(b) Critical Evaluation

Are the Results Realistic?

The reported results are promising but should be interpreted carefully.

The MLP classification accuracy of 86.083% is reasonable for a diabetes classification experiment on the PIMA dataset. However, an accuracy of 86% does not mean that the system is ready for direct clinical diagnosis.

The LSTM result of 87.26% accuracy and correlation of 0.999 appears very strong. However, the unusually high correlation should be interpreted cautiously because correlation measures association between predicted and actual values; it does not by itself establish clinical usefulness.

Therefore, the results demonstrate promising model performance but require independent validation before clinical deployment.

Potential Dataset Bias

The dataset has several limitations:

1. Population Bias

The dataset consists of female patients from the PIMA Indian population.

Therefore, performance may not generalise to:

- Male patients
- Other ethnic/population groups
- Different age distributions
- Different healthcare environments

2. Dataset Size

The dataset contains only 768 records, which is relatively small for developing a healthcare system intended for broad deployment.

3. Class Distribution

The paper reports approximately:

- 65% Healthy
- 35% Diabetic

This indicates an imbalance between the two classes.

4. Limited External Validation

The models are primarily evaluated using the PIMA dataset. A model that performs well on one benchmark dataset may perform differently on real-world hospital data.

Evaluation Protocol Concerns

The use of cross-validation is a positive aspect because it reduces dependence on a single train/test split.

However, stronger evidence would require:

- External validation
- Testing on independent hospital datasets
- Confidence intervals
- Statistical significance tests
- Evaluation across different demographic groups
- Prospective clinical validation

The paper itself highlights the need for further research and broader implementation.

Additional Experiments That Would Strengthen the Study

I would recommend the following experiments:

Experiment 1 – External Dataset Validation

Train the model using PIMA and test it on an independent diabetes dataset.

This would determine whether the model generalises beyond the original dataset.

Experiment 2 – Cross-Population Evaluation

Evaluate the model separately on different demographic groups.

Experiment 3 – Class Imbalance Techniques

Compare:

- SMOTE
- Class weighting

- Random under-sampling
- Original dataset

and determine their effect on Recall and F1-score.

Experiment 4 – Explainable AI

Use techniques such as feature importance or SHAP to explain why a patient is classified as high risk.

Experiment 5 – Clinical Validation

Compare the model's predictions with diagnoses made by qualified medical professionals.

3(c) Real-World Applicability

The proposed system has several potential applications.

1. Remote Diabetes Monitoring

Patients could use wearable/BLE sensors to collect health information and send it to a monitoring system.

2. Mobile Healthcare Applications

A smartphone application could receive patient information and provide monitoring information.

3. Hospital Decision Support

ML predictions could potentially assist healthcare professionals in identifying high-risk patients.

4. Continuous Monitoring

The proposed IoT system could continuously collect measurements such as:

- Blood glucose
- Weight
- Blood pressure
- Demographic information

The paper's proposed architecture uses BLE sensors, an Android mobile device, real-time processing, Kafka and a database/server environment.

Industry Scaling Challenges

1. Data Availability

Healthcare organisations need large, high-quality and representative datasets.

2. Data Privacy

Patient information must be protected from unauthorised access.

3. Integration

ML systems must integrate with existing:

- Hospital databases
- Electronic health records
- Wearable devices
- Clinical workflows

4. Computational Requirements

Real-time prediction requires reliable processing infrastructure.

5. Model Generalisation

A model trained on PIMA data cannot automatically be assumed to work equally well for every population.

6. Ethical Issues

The system must address:

- Patient safety
- Bias
- Transparency
- Privacy
- Accountability
- Human oversight

TASK 4: REFLECTION & LEARNING OUTCOME

4(a) Three Key Insights

Insight 1 – Inductive Learning

I learned that machine learning models can use historical patient records to identify patterns and make predictions for new patients.

The PIMA dataset contains patient attributes such as glucose, blood pressure, BMI, insulin and age, which are used to learn the relationship between medical measurements and diabetes outcomes.

Connection to CO4:
This demonstrates Inductive Learning, where general predictive knowledge is learned from examples.

Insight 2 – Statistical Learning and Classification

The study helped me understand why Logistic Regression is useful for binary classification problems.

The model estimates the probability of a class using the sigmoid function. It provides a relatively simple statistical baseline against which more complex models such as Random Forest and MLP can be compared.

Connection to CO4:
This connects directly to Statistical Learning and classification.

Insight 3 – Model Selection and Evaluation

The article demonstrates that selecting a machine learning algorithm based only on its complexity is not sufficient. Different models perform differently depending on the task.

For classification, the fine-tuned MLP achieved 86.083% accuracy, while for blood-glucose prediction the fine-tuned LSTM achieved 87.26% accuracy.

This shows that model selection should be based on:

- Dataset characteristics
- Problem type
- Evaluation metrics
- Generalisation ability
- Application requirements

Connection to CO4:
This connects to machine learning classification and decision-making.

4(b) Proposed Novel Extension

Proposed Experiment: Explainable and Multi-Dataset Diabetes Prediction Using Ensemble Learning

I would extend this research by developing an explainable ensemble diabetes prediction system using multiple independent diabetes datasets.

Proposed Method

Instead of relying only on the PIMA dataset, I would combine or separately evaluate:

1. PIMA Indian Diabetes Dataset
2. Hospital diabetes dataset
3. Another publicly available diabetes dataset

The models could include:

- Logistic Regression
- Random Forest
- MLP
- LSTM
- Gradient Boosting

The predictions could then be combined using an ensemble approach.

Explainability

An explainability layer could be added to identify the major factors influencing each prediction.

For example:

Patient Prediction: High Diabetes Risk

Main contributing factors:

1. High glucose level
2. High BMI
3. Age
4. Diabetes pedigree function

Why This Is Feasible

This enhancement is feasible because the original study already demonstrates machine learning classification and prediction using multiple algorithms. It also proposes an IoT architecture capable of collecting real-time healthcare information.

The proposed extension would mainly require:

- Additional datasets
- Data standardisation
- Model training
- Cross-dataset testing
- Explainability tools

Expected Impact

The expected benefits are:

- Better generalisation

- Reduced population bias
- More reliable predictions
- Greater clinical transparency
- Better understanding of important diabetes predictors

This would make the research more suitable for real-world healthcare decision-support systems.

OVERALL ARTICLE REVIEW

The selected paper presents a comprehensive machine learning framework for diabetes classification, prediction and IoT-based monitoring. Its major contribution is the combination of multiple ML approaches with a proposed real-time healthcare architecture.

The experimental results show that the fine-tuned MLP achieved 86.083% accuracy, 86.6% precision and 85.1% recall for diabetes classification. For blood-glucose prediction, the fine-tuned LSTM achieved 87.26% accuracy, 0.999 correlation and 2.285 RMSE.

The study is valuable for understanding inductive learning, statistical learning, classification, model comparison and predictive analytics. However, the relatively small PIMA dataset, limited population representation and lack of extensive external clinical validation prevent the reported results from being treated as sufficient evidence for direct clinical deployment.

The proposed IoT architecture is a particularly useful aspect of the paper because it demonstrates how machine learning could eventually be connected with real-time patient monitoring. The architecture incorporates BLE devices, smartphones, real-time processing, Apache Kafka and MongoDB.

Overall, the paper provides a strong example of how Machine Learning + Healthcare + IoT can be combined to support early diabetes identification and monitoring, while also demonstrating the importance of dataset quality, model evaluation and responsible deployment.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand